SPRING 2026

EMBEDDED SYSTEM

MIDTERM REVISION #1

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Theory Questions

1. Write one paragraph about the birth of silicon valley

2. What is the difference between microcontroller and microprocessor

3. State three advantages of using assembly language programming over embedded C. Also state three advantages of using embedded C over assembly language.

4. List the main parts of an embedded system. Also mention different types of embedded processors.

5. Why we chose ATMEL to study? Mention 4 advantages of ATMEL microcontrollers.

6. Give a brief history of AVR microcontrollers.

7. You are trying to program a healthy ATMEGA32 microcontroller but the programming software gives you "Chip Programming Error", what would be the main three reasons for that?

8. Mention 3 different ways to create time delay in AVR C.

9. What is the difference between
   a. ATMEGA16, ATMEGA16L, ATMEGA16A microcontrollers
   b. SBI and SBR instructions; SBIC and SBRC

10. State the different memory types inside AVR microcontrollers and their respective usage. What is a bootloader?

11. Show briefly how car's airbag can be considered an embedded system

12. Mention the main features of ATmega32 ADC

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13. Compare between IN and LDS assembly instructions.

14. What is the difference between microcontroller and FPGA

MCQ Questions

1. What is wrong in the AVR microcontroller

a. 2K RAM
b. 32K flash
c. 10 pins for ADC
d. Cheap

2. Specific Purpose Registers are from data RAM

a. True
b. False

3. Which of the following can match with LDI instruction

a. LDI R34, 12
b. LDI R17, 4
c. LDI R17, 300
d. None

4. We need to read input from port C into R20, which of the following is true

a. IN R20, PORTC
b. IN R20, PINC
c. LDI R20, PORTC
d. LDI R20, PINC

5. Which statement doesn't describe the MACRO

a. The program using macros are more readable
b. MACROs do not use stacks and are not at the risk of stack overflow
c. MACROs take less memory space than subroutine
d. MACRO consists of .MACRO directive, the body, and the .ENDMACRO directive

6. Which code perform the subtraction 15 -12

a) LDI R16, 15
LDI R20, 12
SUB R20, R16
	b) LDI R16, 15
	LDI R20, -12
	SUB R16, R20
		c) LDI R16, 15
		LDI R20, -12
		ADD R16, R20
			c) LDI R16, 15
			SUB R16, 1

7. Which code reads the input from PB2 and put it on PB0

a) CBI DDRB, 2 SBI DDRB, 0 A1: SBIC PINB, 2 RJMP A2 SBI PORTB, 0 RJMP A1 A2: CBI PORTB, 0 RJMP A1	b) CBI DDRB, 2 SBI DDRB, 0 A1: SBIS PINB, 2 RJMP A2 CBI PORTB, 0 RJMP A1 A2: SBI PORTB, 0 RJMP A1	c) CBI DDRB, 2 SBI DDRB, 0 A1: SBIS PINB, 0 RJMP A2 SBI PORTB, 2 RJMP A1 A2: CBI PORTB, 2 RJMP A1	d) CBI DDRB. SBI DDRB. A1: SBIS PINB RJMP A2 SBI PORTI RJMP A1 A2: CBI PORT RJMP A1

8. The operands of MUL instructions are R0 and R1 only

a. True
b. False

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9. The target of a BRNE can be anywhere in the 4M word address space

a. True
b. False

10. In all microcontrollers, the ports have 8 pins

a. True
b. False

11. The first opcode of an AVR program is stored at the address _____ in ROM

a. 0000H
b. Assigned in ORG instruction
c. Assigned by the compiler
d. Available after the interrupt vector

12. Which instruction isn't stored in the memory

a. OUT OCR1B, R3
b. LDI R20, 0XFF
c. ORG 500H
d. STS 300, R1

13. Is the following code is correct

```
EQU MYDATA = 12
LDI R19, MYDATA
ADD R19, MYDATA
STS MYDATA, R19
```

a. True
b. False

14. The ATMEGA 32 doesn't support

a. SPI
b. I2C
c. USART
d. CAN

15. Using the ICALL, the target address of the subroutine is within _____

a. -20248 to +2047
b. -32768 to +32767
c. -2097152 to + 2097151

16. What is the addressing mode in "AND R16, R17"?

a. Register
b. Two Register
c. Immediate
d. Direct

17. Which code reads data from ROM

a)	b)	c)	d)

LDI ZH, HIGH($100)
LDI ZL, LOW($100)
LPM R16. Z
	LDI ZH, HIGH($100<<1)
	LDI ZL, LOW($100<<1)
	LPM R16, Z
		LDI XH, HIGH($100)
		LDI XL, LOW($100)
		LPM R16, X
			LDI XH, [illegible]
			LDI XL, [illegible]
			LPM [illegible]

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Code Questions

1. Write an AVR C program to toggle bits of port B 200 times

```
#include <avr/io.h>
int main (void)
{
    DDRB=0xFF;
    PORTB=0xAA;
    unsigned char z;
    for(z=0;z<200;z++)
     PORTB= ~ PORTB;
while(1);
return 0;
}
```

2. Write an AVR C program to get data from Port C, if it less than 100, send it to Port B; otherwise send it to Port D.

```
#include <avr/io.h>
int main (void)
{
    DDRC =0x00; // to define PC i/p
    PORTC =0xff;// to enable pull up reg
    DDRB=0xff; // to define PB o/p
    DDRD=0xff; // to define PB o/p
    unsigned char temp;
while(1)
  {
      temp=PINC ;
      if(temp<100)

PORTB=temp;
      else

PORTD=temp;
  }
return 0;
}
```

3. Write an assembly program to determine if RAM location 0x200 contains the value 0. If so put 0x55 in to it

```
LDS R30, 0X200
CPI R30, 0
BRNE NEXT
LDI R30, 0X55
STS 0X200, R30
NEXT: RJMP NEXT
```


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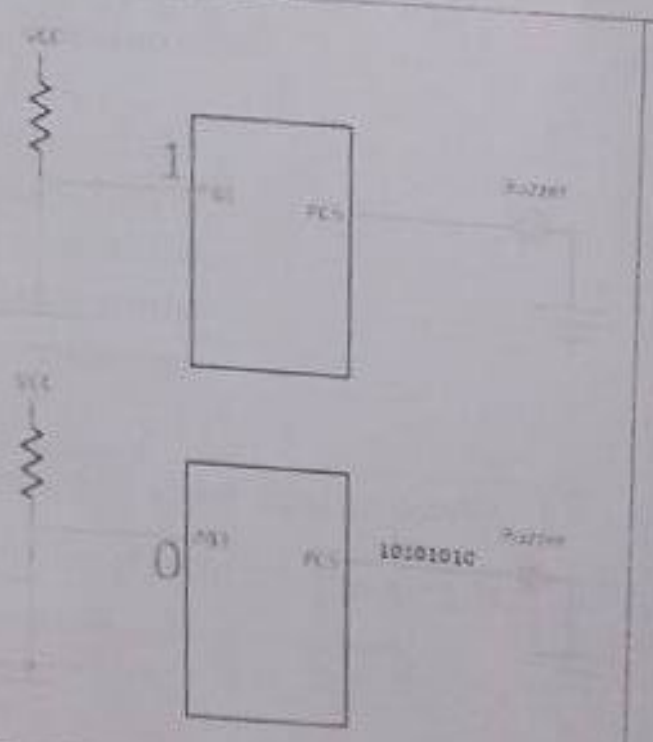
4. Write a program to monitor PORTB continuously until it has 63H value

```
        LDI R20, 0X00
        OUT DDRB, R20
        LDI R21, 0X63
AGAIN:  IN R20, PINB
        CP R20, R21
        BRNE AGAIN
```

5. Write a program that adds the content of location 0x220 to location 0x221 and save result to 0x221

6. Door alarm switch is connected to PB3 when door is open PB3 receives 0. At this condition turn on the buzzer connected to PC5 by sending high to low signal

```
        CBI DDRB,3
        SBI DDRC,5
HERE:   SBIC PINB,3
        RJMP HERE
        SBI PORTC,5
        CBI PORTC,5
```



7. Write an assembly and C language program to generate 50% duty cycle square wave with the highest possible frequency on bit 5 of PORTB.

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8. Assume R20 has packed BCD 29. Write a program to convert it in to equivalent ASCII and store the result in R21 and R22

```
.INCLUDE "M32DEF.INC"
LDI R20, 29
MOV R21, R20
ANDI R21, 0X0F
ORI R21, 0X30
MOV  R22, R20
SWAP R22
ANDI R22, 0X0F
ORI R22, 0X30

HERE: JMP HERE
```

9. Write an AVR assembly language program to copy a block of 16 bytes of data from data memory location starting 0x130 to data memory locations starting at address 0x60

```
      LDI R16,16
      LDI XL,0X30
      LDI XH,0X01
      LDI YL,0X60
      LDI YH,0X00
L1:   LD R20,X+
      ST Y+,R20
      DEC R16
      BRNE L1
```

10. A push button is connected to PB.0 and a LED is connected to PC.0. Write an AVR assembly language to turn on the LED when the push button is pressed low and then turn it OFF.

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11.

d) Write an assembly language macro called LOAD which loads an immediate value into an IO register. Show how to use this macro.

12.

f) Write an assembly language program to transfer a string of data from flash address 0x1000 to RAM starting from address 0x200.

13. Write an assembly code to multiply the contents of two memory locations whose addresses are 0x100 and 0x150 and store the result in memory location 0x200 for least significant byte and 0x201 for most

General Questions

1. Assuming that r16 contains the value 0x33. What value would be in r16 after the instruction cpi r16, 0x33 is executed?
2. Assuming that r16 contains the value 0x33. What value would be in r16 after the instruction subi r16, 0x33 is executed?
3. Assuming that r16 contains the value 0x33. What 1-bit value would be contained in the Z-bit in SREG after the instruction cpi r16, 0x33 is executed?
4. Using just two instructions, test the contents of register16, if all the bits are cleared (equal to zero) branch to the label is_zero.
5. Write an instruction sequence to toggle bit 4, 2, and 0 in register 16.
6. Write an instruction to clear bit 4, 2, and 0 in register r16 without using the cbr instruction.
7. What is the instruction equivalent to this hexa value (Hexa of LDI is E) E205

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8. Write down Single C and assembly statement to do the following

   a. Toggle all bits of PORTC

   b. Wait forever

   c. Set bit PD5 as output

Theory Questions Answers

1. Birth of Silicon Valley

Silicon Valley originated in the southern part of San Francisco Bay Area during the mid-20th century, driven by research from Stanford University and the growth of semiconductor companies. Early pioneers like Fairchild Semiconductor and Intel helped establish the region as a global hub for electronics and innovation, particularly in silicon-based integrated circuits, which gave the area its name.

Hewlett Packard (HP) is the world's largest information Technology Company. Its foundation is called the Birthplace of Silicon Valley. On a quiet, tree-lined residential street near Stanford University, the HP Garage stands today as the enduring symbol of innovation and the entrepreneurial spirit.

2. Microcontroller vs Microprocessor

Feature	Microcontroller	Microprocessor
Definition	CPU + memory + peripherals on one chip	CPU only
Usage	Embedded systems	General-purpose computing
Cost	Low	Higher
Power consumption	Low	Higher
Example	ATmega32	Intel 8086

3. Assembly vs Embedded C

<u>Advantages of Assembly</u>

1. Faster execution (optimized instructions)
2. Direct hardware control

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3. Smaller code size

Advantages of Embedded C

1. Easier to write and maintain
2. Portable across platforms
3. Faster development time

4. Main Parts of Embedded System Components:

- Microcontroller / Processor
- Memory
- Input/Output devices
- Power supply
- Software (firmware)

Types of Embedded Processors:

- Microcontrollers (e.g., AVR)
- Microprocessors
- DSP (Digital Signal Processors)
- ASIC
- FPGA

5. Why ATMEL?

Advantages:

1. Best Flash memory in the world
2. Low cost
3. Easy to program
4. Rich peripherals (ADC, timers, etc.)
5. Variety in packages (from 6 pins to 128 pins)
6. Efficient RISC architecture

6. Brief History of AVR

AVR is designed by two Norwegian students Alf-Egil Bogen and Vegard Wollan. AVR microcontrollers were then developed by Atmel in the 1990s. The architecture was designed as a **RISC-based system** for high performance and low power consumption. Later, Atmel was acquired by Microchip Technology, which continues AVR development.

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7. "Chip Programming Error" Reasons

1. Wrong wiring (MOSI, MISO, SCK, RESET)
2. Incorrect clock source / fuse settings
3. No power or unstable supply

8. Ways to Create Delay in AVR C

1. Using _delay_ms() / _delay_us()
2. Using timer interrupts
3. Using software loop

9. Differences

(a) ATMEGA16 vs ATMEGA16L vs ATMEGA16A

Type	Difference
ATmega16	Standard version
ATmega16L	Low voltage version
ATmega16A	Improved power and performance

(b) Instructions

Instruction	Meaning
SBI	Set bit in I/O register
SBR	Set bit in general register
SBIC	Skip if bit in I/O register is clear
SBRC	Skip if bit in register is clear

10. Memory Types in AVR

Memory	Usage
Flash (ROM)	Program storage
SRAM	Temporary data

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Memory	Usage
EEPROM	Non-volatile data

Bootloader: A bootloader is a small program stored in Flash that allows programming the microcontroller without external programmer (via UART, etc.).

11. Airbag as Embedded System

A car airbag system uses:

- Sensors (impact detection)
- Microcontroller
- Actuator (airbag deployment)

When a collision is detected, the microcontroller processes data and triggers the airbag instantly → real-time embedded system.

12. Features of ATmega32 ADC

- 10-bit resolution
- 8 input channels
- Successive approximation method
- Adjustable reference voltage
- Interrupt support

13. IN vs LDS

Instruction	Description
IN	Reads from I/O register (fast, limited range, takes less memory space)
LDS	Loads from SRAM (full memory access, slower, takes more memory space)

14. Microcontroller vs FPGA

Feature	Microcontroller	FPGA
Structure	Fixed architecture	Programmable hardware

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Feature	Microcontroller	FPGA
Flexibility	Limited	Very high
Speed	Moderate	Very high
Design	Software-based	Hardware-based (HDL)

MCQ Answers

1. C
2. A
3. B
4. B
5. C
6. C
7. D
8. B
9. B
10. B
11. A
12. C
13. B
14. D
15. B
16. B
17. B

General Questions Answers

1. R16 = 0X33
2. R16 = 0
3. Z = 1
4. TST R16
   BREQ is_zero
8.
   a. PORTC ^= 0xFF;
   Assembly
   LDI R16, 0xFF
   IN R17, PORTC
   EOR R17, R16
   OUT PORTC, R17
   b. while(1) , HERE: RJMP HERE
   c. DDRD |= (1<<PD5); SBI DDRD, PD5
5. LDI R17, 0x15
   EOR R16, R17
6. ANDI R16, 0Xea
7. LDI R16, 0x25

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Register indirect with displacement

```
LDD    Rd, Z+q        ;load from Z+q into Rd
```

where q is a number between 0 to 63, and Rd is any of the general purpose

```
STD    Z+q,Rr         ;store Rr into location Z-q
```

For example, the following instruction writes the contents of R20 into the location that is five bytes away from where Z points to:

```
STD    Z+5,R20        ;store R20 into location Z+5
```

Flash Indirect Addressing mode

Load data from **ROM** to **RAM**

Syntax:	Operands:	
(i) LPM	None	Store in R0 by default from pointer Z
(ii) LPM Rd, Z	$0 \le d \le 31$	
(iii) LPM Rd, Z+	$0 \le d \le 31$	

Data Transfer from ROM to RAM with address 0x140 onwards

```
        .ORG 0
        .INCLUDE "M32DEF.INC"
        LDI ZH,HIGH(MYDATA<<1)
        LDI ZL,LOW(MYDATA<<1)
        LDI XH,0X01
        LDI XL,0X40
L1:     LPM R20, Z+
        CPI R20,0
        BREQ HERE
        ST X+,R20
        RJMP L1
HERE:   RJMP HERE

MYDATA: .DB "THE WORLD PEACE",0
```


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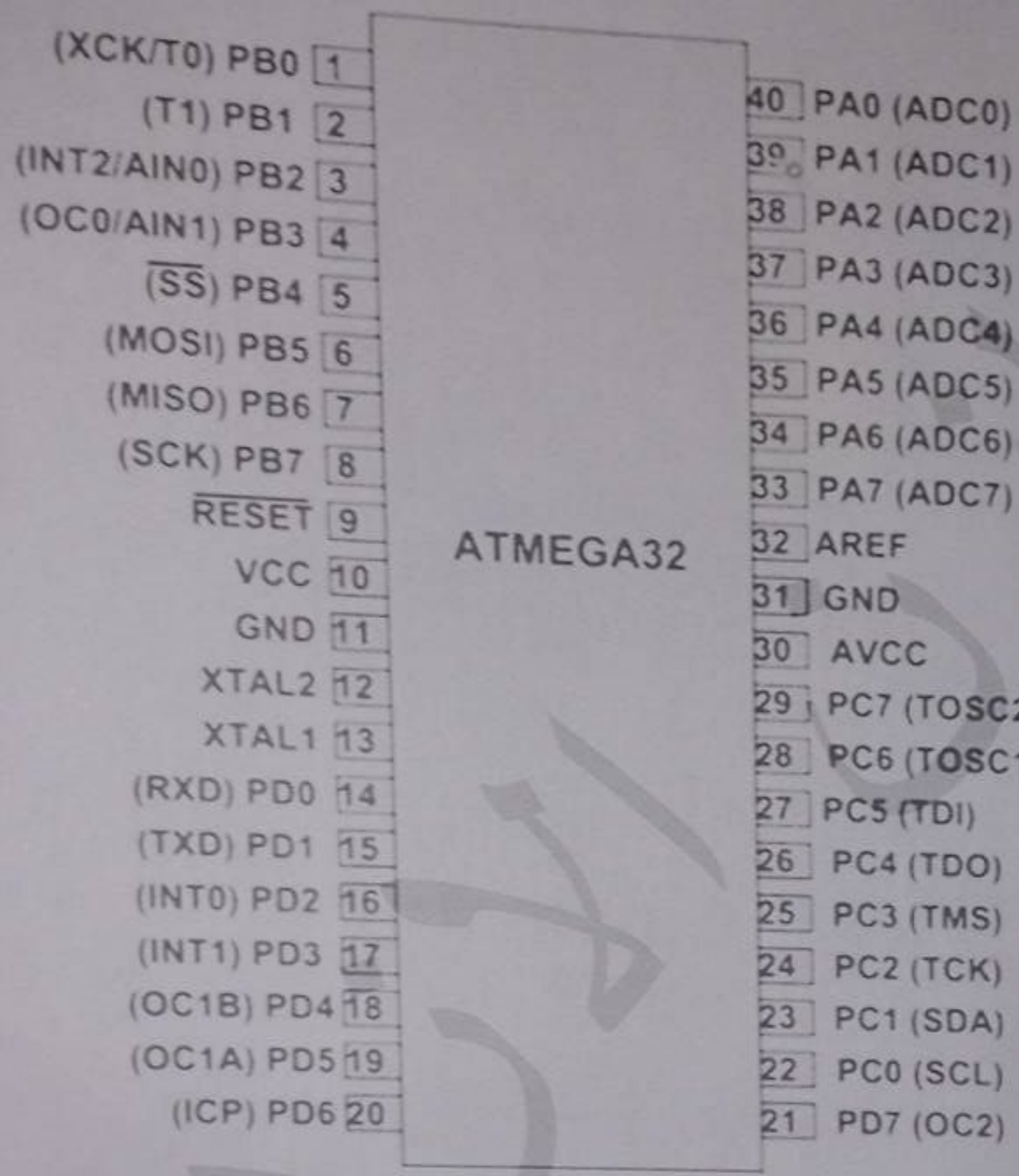
ATMEGA32
(XCK/T0) PB0 1
(T1) PB1 2
(INT2/AIN0) PB2 3
(OC0/AIN1) PB3 4
(SS) PB4 5
(MOSI) PB5 6
(MISO) PB6 7
(SCK) PB7 8
RESET 9
VCC 10
GND 11
XTAL2 12
XTAL1 13
(RXD) PD0 14
(TXD) PD1 15
(INT0) PD2 16
(INT1) PD3 17
(OC1B) PD4 18
(OC1A) PD5 19
(ICP) PD6 20
40 PA0 (ADC0)
39 PA1 (ADC1)
38 PA2 (ADC2)
37 PA3 (ADC3)
36 PA4 (ADC4)
35 PA5 (ADC5)
34 PA6 (ADC6)
33 PA7 (ADC7)
32 AREF
31 GND
30 AVCC
29 PC7 (TOSC2)
28 PC6 (TOSC1)
27 PC5 (TDI)
26 PC4 (TDO)
25 PC3 (TMS)
24 PC2 (TCK)
23 PC1 (SDA)
22 PC0 (SCL)
21 PD7 (OC2)

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